

Unit 1: Asymmetric Cryptography

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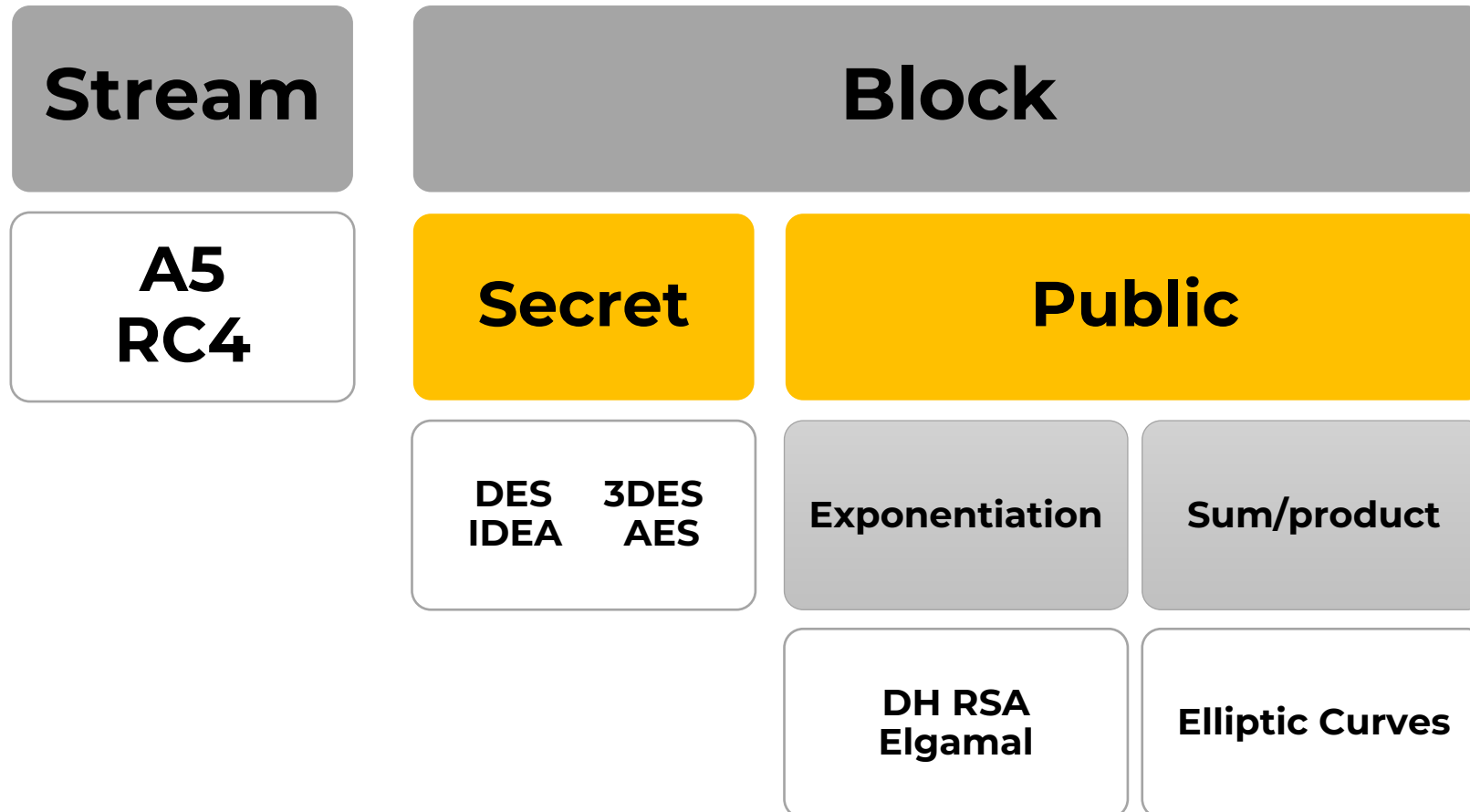
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Introduction

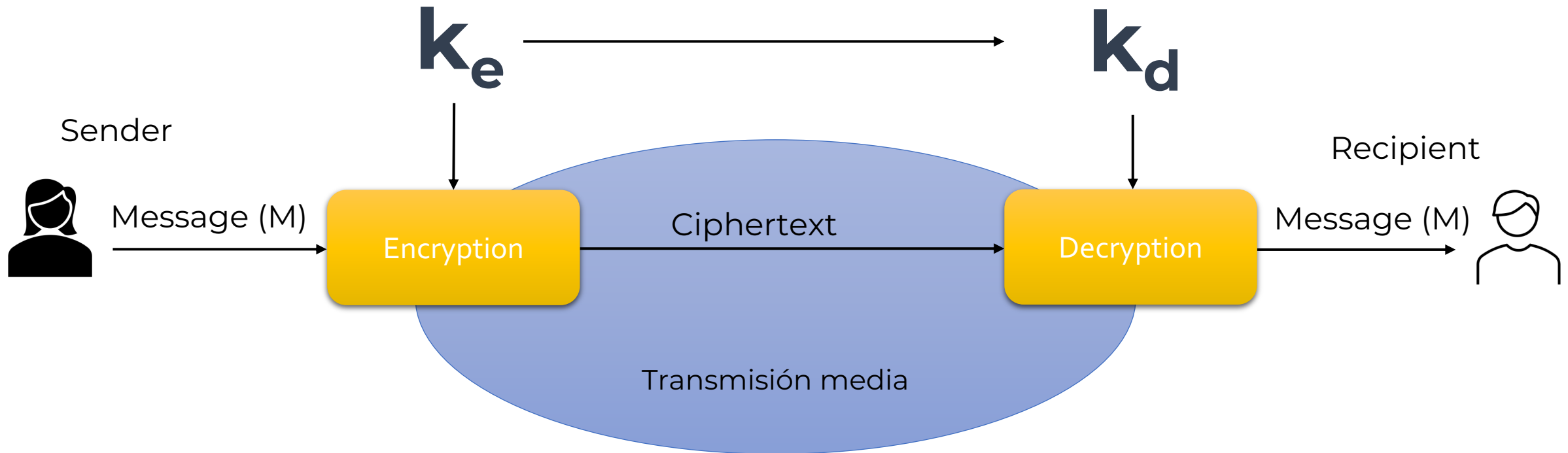
Types of Encryption

- **Depending on the keys**
 1. **Symmetric (or secret key):** Same key used for both operations
 2. **Asymmetric (or public key):** One key encrypts the other decrypts
- **Depending on how encryption works**
 1. **Stream cipher:** encryption is done bit by bit
 2. **Block cipher:** encryption is done in blocks

Modern Ciphers



Asymmetric Encryption



Asymmetric Encryption

- We use 2 keys
 - Public key (**e** from *encrypt*)
 - Private key (**d** from *decrypt*)
 - Both keys are inverse within a finite field
 - One encrypts, the other decrypts (and the other way around)
- Security
 - Relies on the computational complexity of obtaining one key (private one) from the public key without knowing an additional secret (we call trap) o
- Allows us to send encrypted messages without sharing a key via another channel (as in symmetric cryptography)

Keys

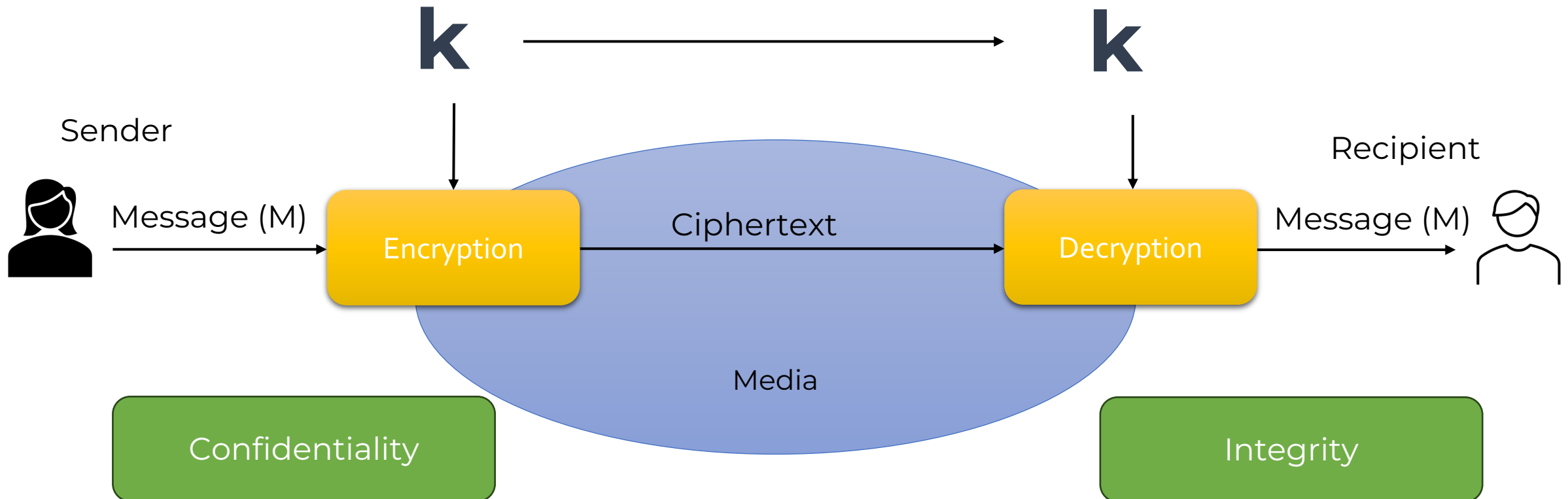
- Asymmetric key algorithms use 2 keys
- Public
 - Allows us to encrypt information to send it to an entity
 - We use the public key of the recipient
- Private
 - Allows the recipient to decrypt information they have been sent

In Practice

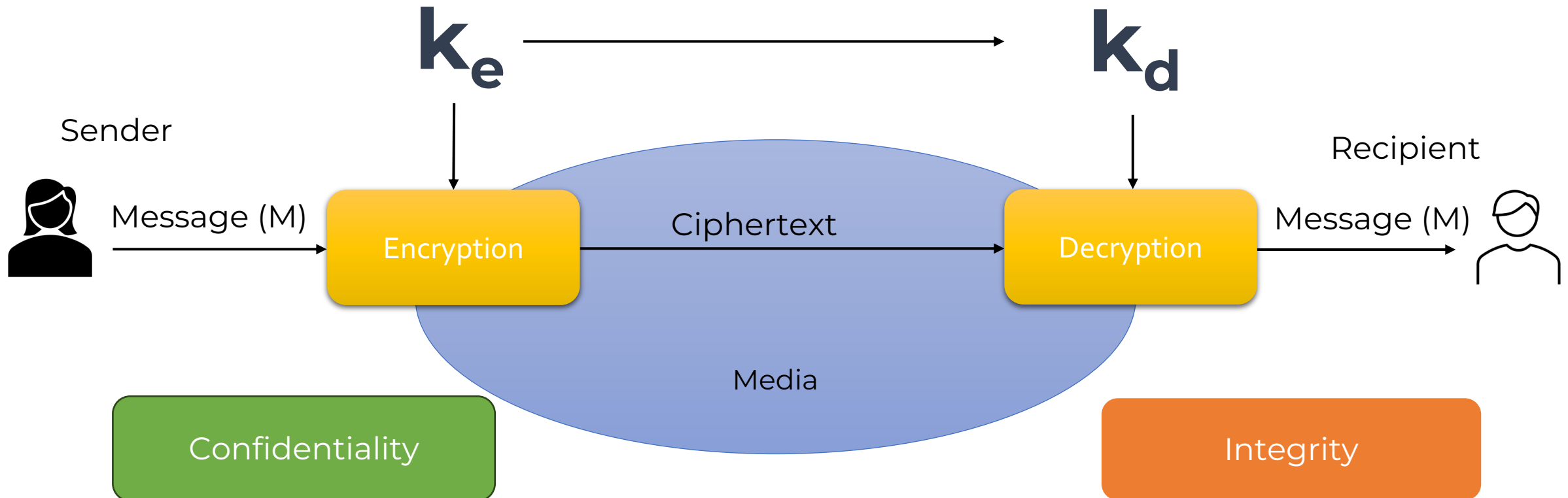
- Two main methods
 - Exponentiation
 - $C = A^B \bmod n$
 - Elliptic Curves
 - Unit 4

Symmetric vs Asymmetric Cryptography

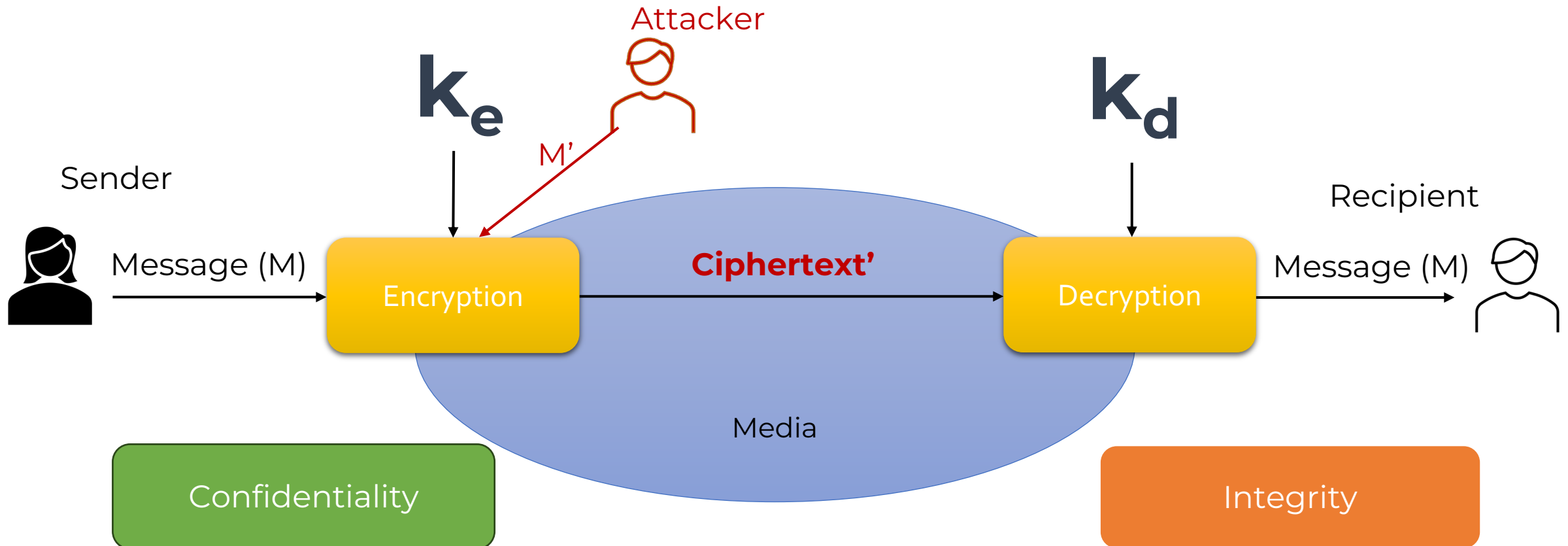
Confidentiality and Integrity in Symmetric Cryptography



Confidentiality and Integrity in Asymmetric Cryptography



Confidentiality and Integrity in Asymmetric Cryptography



¿How do we address integrity in asymmetric cryptography?

- Digital Signature
- We will use our private key to encrypt something that is already known
- Can also be decrypted with our public key so the receiver can be sure we were the only ones capable of encrypting it
- More on this during units 5 and 6

Encryption security

Symmetric Cryptography

- One Key
- Security relies on the key
 - Keep the key secret
 - Difficult to guess
 - As random as possible
 - As long as possible

Asymmetric Cryptography

- Two keys
- Security relies on difficulty of getting one key from the other
 - Depends on the problem
 - Exponentiation
 - Elliptic curves

Key Management

Symmetric Cryptography

- Each communication requires a different key
- Keys need to be exchanged via another secure channel

Asymmetric Cryptography

- Two keys per user
- Public key is shared
- Private key is kept secret

How many keys do we need for n users?

$$\frac{n(n-1)}{2}$$

$$2n$$

Other considerations

	Symmetric	Asymmetric
Types	Flujo y bloque	Bloque
Algorithms	RC4, DES, AES	RSA, Curvas Elípticas
Key size	128 bits, 256 bits	1024 bits, 2048 bits, 256 bits (EC)
Number of keys	$n(n-1)/2$	$2n$
Key duration (time)	Short	Long
Encryption speed	High	Slow
Confidentiality	Yes	Yes
Integrity	Yes	No
Susceptible to Man in The Middle	No	Yes
Allows key exchange	No	Yes

Hybrid systems

- Asymmetric Cryptography allows us to exchange information without the need of a previously secured channel
- We use asymmetric cryptography to share the key we will use during symmetric encryption
- We take advantage of the security of asymmetric cryptography and the speed of symmetric cryptography

Diffie-Hellman Key Exchange

Diffie Hellman

- November 1976 Stanford
- One of the first public key systems
- Allows to establish a shared key between two parties with no previous information exchange via another secure channel
- Based on trap functions
 - $x \rightarrow f(x)$ is easy to calculate
 - $f(x) \rightarrow x$ is very computationally impossible without knowing t (the trap)
 - DH uses the discrete logarithm problem

The Discrete Logarithm Problem

- Modular exponentiation
 - We pick a large prime **p** , and **α** , a generator within the finite field of **p** . Both values will be public
 - A generator is an element capable of generating all values between 1 and $p-1$ when it is exponentiated within that field
 - $\alpha^1 \bmod p$,
 - $\alpha^2 \bmod p$,
 - ...
 - $\alpha^{p-1} \bmod p$, provide all values between 1 and $p-1$
 - With a secret value **x** we calculate $y = \alpha^x \bmod p$
- The discrete logarithm problem
 - Knowing **y** , **α** and **p** it is computationally unfeasible to obtain $x = \log_{\alpha} y \bmod p$

Generators

- A generator is an element capable of generating all values between 1 and $p-1$ when it is exponentiated within that field
 - $\alpha^1 \bmod p$
 - $\alpha^2 \bmod p$
 - ...
 - $\alpha^{p-1} \bmod p$
 - Result in all values between 1 and $p-1$

$$2^1 \bmod 7 = 2 \bmod 7 = 2$$

$$2^2 \bmod 7 = 4 \bmod 7 = 4$$

$$2^3 \bmod 7 = 8 \bmod 7 = 1$$

$$2^4 \bmod 7 = 16 \bmod 7 = 2$$

$$2^5 \bmod 7 = 32 \bmod 7 = 4$$

$$2^6 \bmod 7 = 64 \bmod 7 = 1$$

Which of these two (2 or 3) is a generator within the field $p=7$?

$$3^1 \bmod 7 = 3 \bmod 7 = 3$$

$$3^2 \bmod 7 = 9 \bmod 7 = 2$$

$$3^3 \bmod 7 = 27 \bmod 7 = 6$$

$$3^4 \bmod 7 = 81 \bmod 7 = 4$$

$$3^5 \bmod 7 = 243 \bmod 7 = 5$$

$$3^6 \bmod 7 = 729 \bmod 7 = 1$$

Diffie Hellman Algorithm

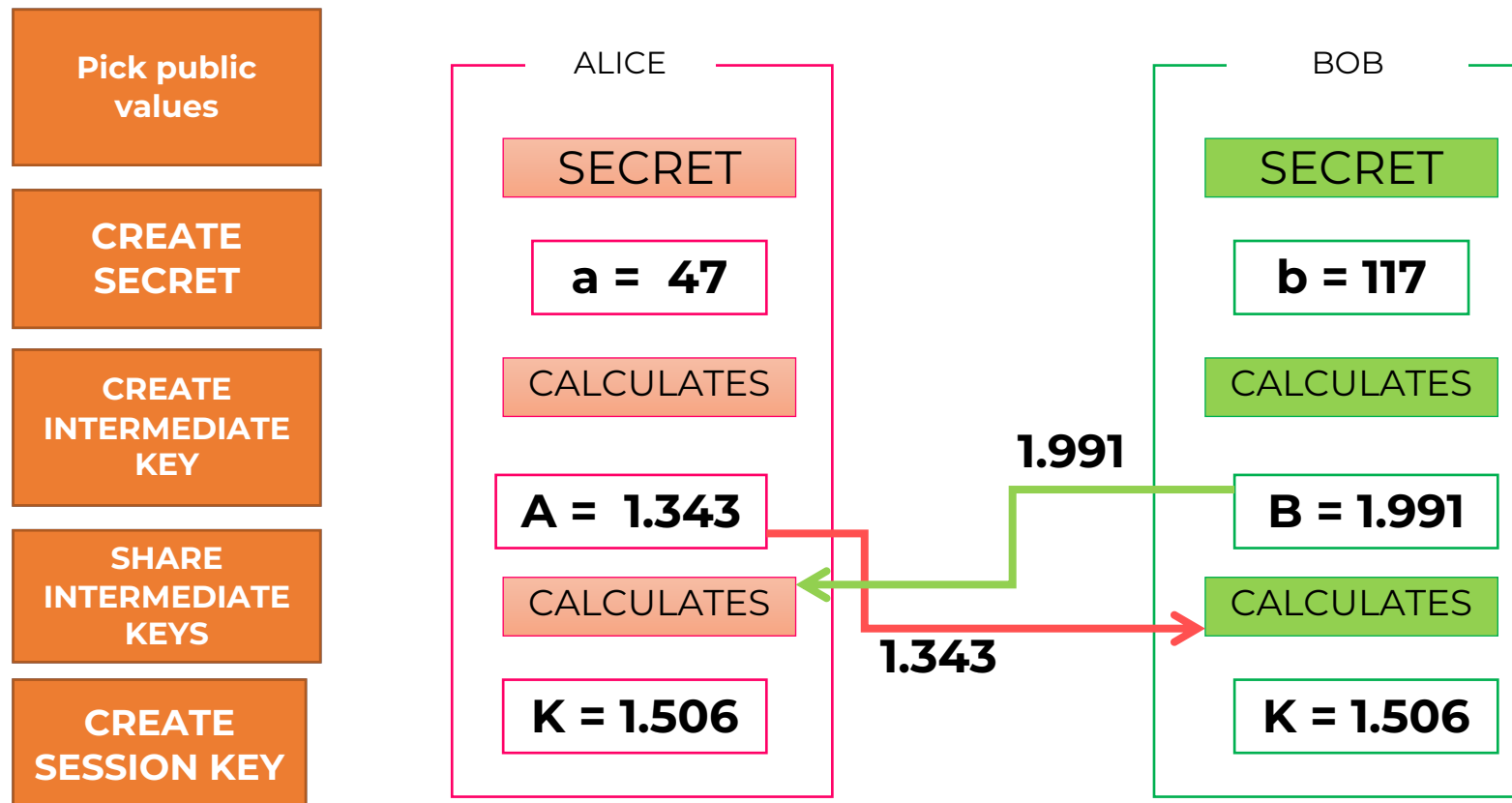
- A generates a random number **a** and sends B: $\alpha^a \bmod p$
- B generates a random number **b** and sends A: $\alpha^b \bmod p$
- B uses **b** and calculates $(\alpha^a)^b \bmod p = \alpha^{ab} \pmod{p}$
- A uses **a** and calculates $(\alpha^b)^a \bmod p = \alpha^{ba} \pmod{p}$

The shared secret, between A and B will be $\alpha^{ab} \bmod p$

DH in steps

Prime and modulo $\rightarrow p = 1.999$

Generator of $p \rightarrow \alpha = 33$



Steps in DH (2)

- Alice (A) and Bob (B) establish two public values the prime $p = 1.999$, and the generator $\alpha = 33$.
- Alice picks/generates the secret $a = 47$
- Bob picks/generates the secret $b = 117$
- Steps
 1. A calculates $\alpha^a \bmod p = 33^{47} \bmod 1.999 = 1.343$ and sends it to B.
 2. B calculates $\alpha^b \bmod p = 33^{117} \bmod 1.999 = 1.991$ and sends it to A.
 3. B receives 1.343 and calculates $1.343^{117} \bmod 1.999 = 1.506$.
 4. A receives 1.991 and calculates $1.991^{47} \bmod 1.999 = 1.506$.
 5. The value $\alpha^{ab} \bmod p = 33^{47 \cdot 117} \bmod 1.999 = \mathbf{1.506}$
 6. The shared key k is $33^{5.499} \bmod 1.999 = \mathbf{1.506}$

Attacks

- p and α are public
- If we intercept $\alpha^a \bmod p$ or $\alpha^b \bmod p$
 - ¿Can we calculate $\alpha^{ab} \bmod p$?
 - Brute force
 - Solving the discrete logarithm problem
- ¿Are there any other ways of attacking this?
 - Man in the Middle

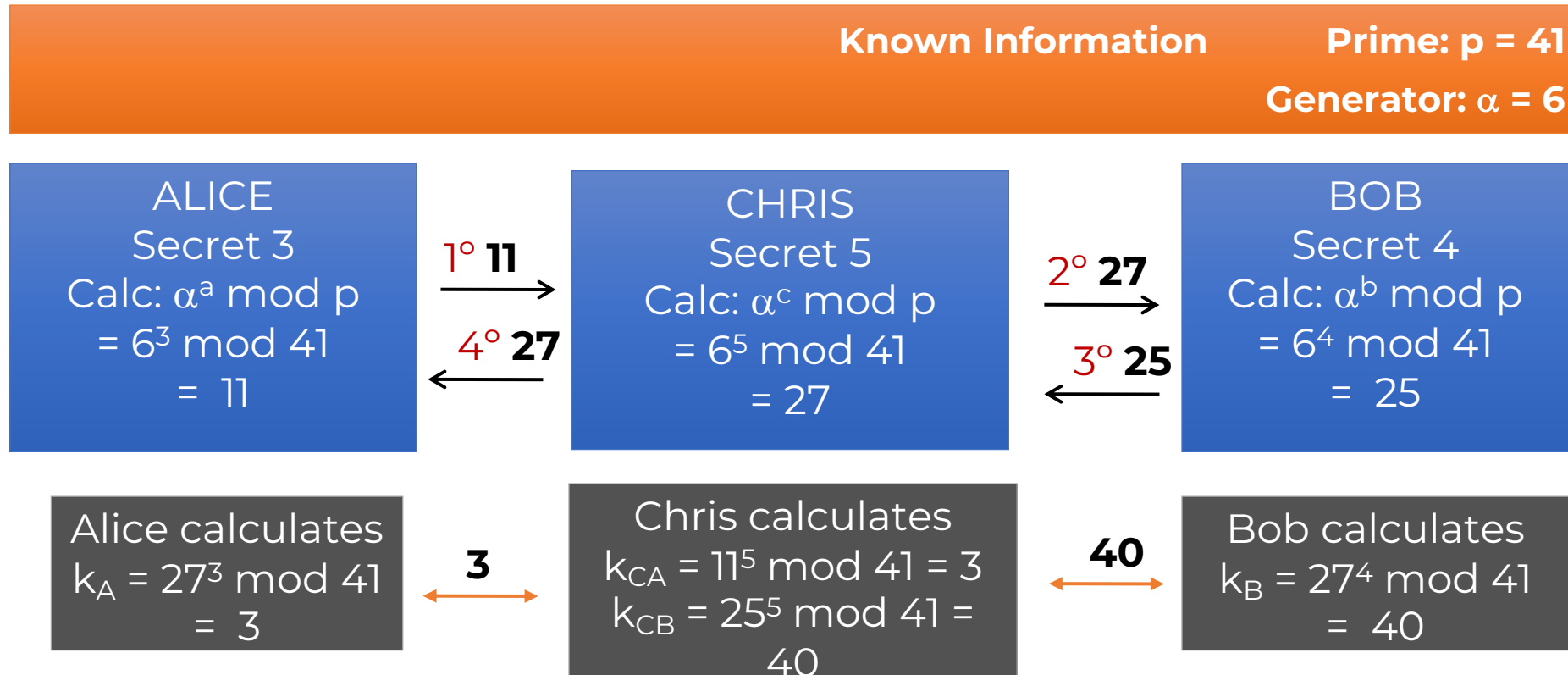
Brute force attack

- Attacker knows
 - Values of $\alpha = 33$ and $p = 1.999$ (public)
 - The value Alice sends Bob= 1.343 (has access to channel)
 - The value Bob sends Alice= 1.991 (has access to channel)
- They can calculate a and b by solving
 - **$33^{x_a} \bmod 1.999 = 1.343$** for all possible values of x_a
 - **$33^{x_b} \bmod 1.999 = 1.991$** for all possible values of x_b
- In the real-world these numbers are so large so this problem is too complex for a computer to solve

Attacking the discrete logarithm problem

- If $C = \alpha^a \bmod p$
 - The **attacker** will try to do the inverse operation: $a = \log_{\alpha} C \bmod p$.
- We don't know of any efficient algorithm that can solve this problem in polynomial time
 - If we increase the size of the input, the time it requires to be solved won't grow lineally
- Computationally intractable for large values of p (1.000 bits)

Man in the Middle



Fast exponentiation

Issues when doing exponentiations

- Calculating C with $N = 48$ and $\text{exponent} = 5$ and $n = 91$:

- $C = N^e \bmod n = 48^5 \bmod 91 = 254.803.968 \bmod 91 = 55$
- We can do $48^5 = 254.803.968$ because e is small

- Calculating N with C, $e = 29$ y $n = 91$:

- $N = C^d \bmod n = 55^{29} \bmod 91 = 48$ (we recover the secret = 48)
- But 55^{29} is very large (51 digits):

$$55^{29} =$$

295.473.131.755.644.748.809.642.476.009.391.248.226.165.771.484.375

- ¿Can we do this with a calculator?

Summary

Summary

- Asymmetric cryptography uses two keys
 - One encrypts the other one decrypts
- Symmetric vs Asymmetric crypto
 - There are relevant differences between the two that affect how they are used in the real world
- Diffie-Hellman
 - Allows us to establish a shared secret
 - Susceptible to Man in the Middle attacks
- Exponentiation

Questions?

